

# DVA Assignment 1: Viva & Concept Notes

A quick overview of concepts for teacher questions

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## Topic 1: Data Wrangling (Canada Dataset)

### Q: What is Data Wrangling?

Data wrangling is the process of cleaning, structuring, and enriching raw data into a desired format for better decision making. In this assignment, we manipulated rows/columns to extract meaningful insights.

### Key Actions Taken:

- Added a 'Total' column by summing immigrants from 1980 to 2013.
- Checked for Null Values (found 0 missing values).
- Sorted data to find the Top 5 countries (India, China, UK, Philippines, Pakistan).

## Topic 2: Data Preprocessing (Laptop Dataset)

### Q: How did you handle missing data?

We had missing values in RAM, Weight, and GPU. For RAM and Weight (numerical), we used the **Median** instead of Mean because Median is robust and not affected by extreme outliers. For GPU (categorical), we filled missing data with the placeholder 'Unknown'.

### Q: What is the difference between Standardization and Normalization?

- **Standardization (Z-score):** Transforms data to have a mean of 0 and standard deviation of 1. Good for algorithms that assume Gaussian (normal) distribution.
- **Normalization (Min-Max Scaling):** Compresses data into a fixed range, usually between 0 and 1. Good when the data doesn't follow a normal distribution.

### Q: What is Label Encoding?

Machine learning models cannot read text. Label Encoding converts categorical text data (like Company names 'Dell', 'HP') into numerical values (like 0, 1).

## Topic 3: Hypothesis Testing (Insurance Dataset)

### Q: What does a p-value mean?

The p-value tells us if our result is statistically significant. If **p-value < 0.05**, we Reject the Null Hypothesis ( $H_0$ ). If **p-value > 0.05**, we Accept the Null Hypothesis.

### Q: Explain the different Statistical Tests used.

**1. T-Test:** Used to compare the means of exactly TWO groups.

- Example: Comparing BMI of Males vs Females. (Result: No difference,  $p > 0.05$ )

- Example: Comparing Medical Charges of Smokers vs Non-smokers. (Result: Smokers pay more,  $p < 0.05$ )

**2. Pearson Correlation:** Used to check the linear relationship between two continuous variables.

- Example: Age vs Charges. As age increases, do charges increase? (Result: Yes, positive correlation,  $p < 0.05$ )

**3. Chi-Square Test:** Used to check relationship between two CATEGORICAL variables.

- Example: Region vs Smoking Habit. Are some regions more likely to smoke? (Result: No, independent,  $p > 0.05$ )

**4. ANOVA (Analysis of Variance):** Used to compare the means of THREE OR MORE groups.

- Example: Comparing BMI among women with 0, 1, or 2 children. (Result: No significant difference,  $p > 0.05$ )

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## Topic 4: Data Visualization Concepts

### Q: When do you use different types of plots?

- **Line Plot:** Best for showing trends over time (e.g., immigration from 1980-2013).

- **Box Plot:** Best for showing data distribution, quartiles, and identifying outliers.

- **Scatter Plot:** Best for showing the relationship or correlation between two numerical variables.

- **Histogram:** Best for showing frequency distributions (e.g., how many countries sent X number of immigrants).

- **Grouped Bar Chart:** Best for comparing sub-categories across main categories (Segmentation).

*End of Notes - Best of luck for the submission and viva!*